STAFF REVIEW · H446 245 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. max(3,6)+2=8 minutes.

2. Test-driven development or continuous integration.

3. To identify missing needs, usability issues or changes before more costly development.

4. Factor ¬A: F=¬A∧(¬B∨B)=¬A∧1=¬A. This also forms a pair of adjacent 1s on a K-map.

I Do — source worked steps

1. Predict the method — Recommend Spiral because technical risks can be evaluated at each iteration.

2. Check and explain — Explain the disadvantage: risk analysis adds project management and cost.

We Do — guided CHECK

1. Predict the method
RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.
Check and explain
Qualify the recommendation: frequent prototyping must not replace security and validation testing.

You Do — source mark schemes

1. [4] Waterfall sequential stages [1]; Spiral risk-driven iterations [1]; RAD rapid prototyping with users [1]; XP practices such as TDD/pair programming [1].

2. [6] Spiral [1]; explicit risk analysis [1]; repeated validation [1]; manage technical uncertainty [1]; formal review [1]; greater cost/overhead [1].

3. [6] Agile permits feedback and changed UI requirements [2]; Waterfall offers well-defined approval/documentation milestones [2]; propose phased/hybrid governance or justify one method [1]; discuss risk of late change or fixed-scope rigidity [1].

Exit Ticket — source answer

1. False. Project constraints determine the appropriate approach.